



# Project

## MIRROR

An AI Powered Memory and Emotional Analytics System

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First Review  
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# Outline

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# Problem Recap

## **Core Problem:**

Human brains are built to move on, causing "Aha!" moments and life lessons to fade within days.

## **The "Fade":**

You might have a 2:00 AM insight about a career or relationship, but the next afternoon, the urgency of that feeling is gone.

## **Passive Silos:**

Tools like Notion, Keep or WhatsApp are "Static Systems" where information is recorded but stays isolated and forgotten.

## **Contextual Gap:**

Current AI tools lack long term personal context, they don't connect a lesson from months ago to a struggle today.

## **Lack of Reinforcement:**

There is no system designed to proactively "mirror" past breakthroughs back to the user.



# Literature Review

## Theme 1: Emotional Tagging & Long Term Memory (LTM)

**Core Insight:** Emotional AI systems that recognize human feelings in real time significantly improve information retention and user motivation.

**Citation:** (Deckker & Sumanasekara, 2025 – *Scopus Indexed*)

## Theme 2: Semantic Stability in Memory Augmented LLMs

**Core Insight:** Standard AI suffers from "Semantic Decay" (loss of nuance over time). Persistent context is required to maintain a stable "life story" for the user.

**Citation:** (Smith, 2025 – *IEEE/Medium*)

## Theme 3: Proactive Behavioral Mirroring

**Core Insight:** Personalized, proactive AI interventions create measurable benefits in resilience and prevent "mindfulness decline" compared to traditional static notes.

**Citation:** (Cachia et al., 2025 – *HBS Research*)

## Theme 4: Vector Storage for Contextual Coherence

**Core Insight:** Utilizing Vector Databases (like pgvector) can increase task success rates by 10–20% by allowing the AI to selectively remember past breakthroughs.

**Citation:** (VIMBank Research, 2024 – *ResearchGate/IJ*)



# Theoretical Background

## Cognitive Psychology: Emotional Tagging

**The Theory:** The brain remembers emotional events better than neutral ones. (Deckker & Sumanasekara, 2025).

**MIRROR Application:** The system "tags" reflections with feelings. This ensures that a big personal breakthrough isn't "buried" under daily routine notes.

## Information Science: Semantic Vector Space

**The Theory:** Turning text into numbers (Vectors) based on **meaning**, not just keywords. (VIMBank Research, 2024).

**MIRROR Application:** Using math to find "distance" between entries. If you have a problem today, the AI mathematically finds a lesson you learned months ago that "matches" the current situation.

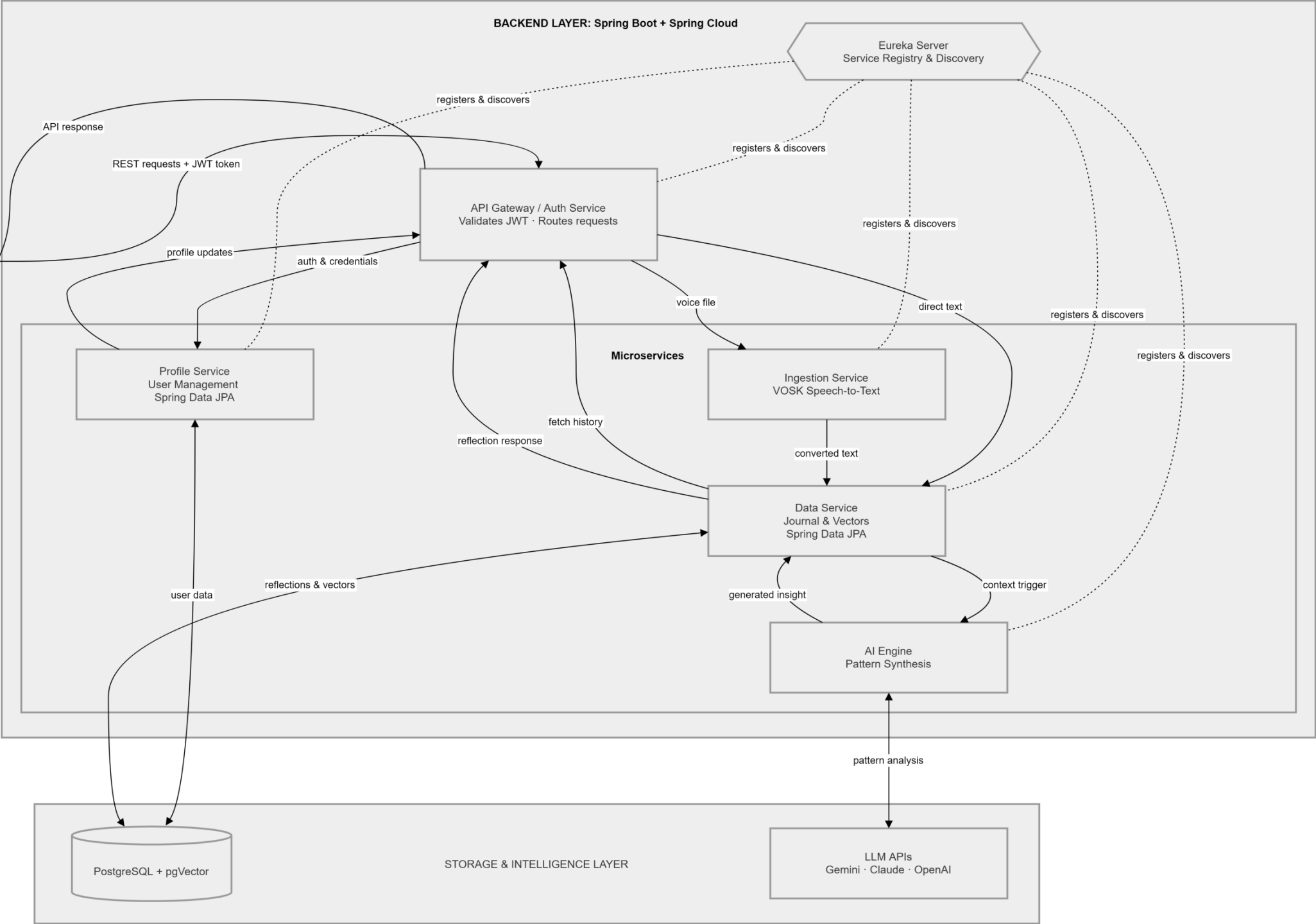
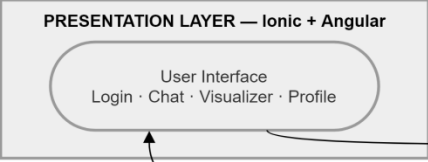
## Computing Model: Retrieval-Augmented Generation (RAG)

**The Theory:** Standard AI "forgets" long conversations. RAG connects the AI to a permanent database (Memory). (Smith, 2025).

**MIRROR Application:** Instead of the AI starting "fresh" every time, it looks into your **pgvector** database to remember your whole life story before it gives you advice.



# Proposed System Architecture





# Research Design

## Research Design:

### Simulation Based Experimental Design:

Utilizing synthetic user personas and history to validate the pattern matching accuracy of the MIRROR engine without compromising real user privacy during the alpha phase.

## Sampling & Validation Strategy:

### Phase 1 (Automated Validation):

Deployment of a **Synthetic Dataset (100+ generated 'Aha!' moments)** to stress test the pgvector search.

**Goal:** Measuring **Recall @ K** to evaluate how effectively the AI retrieves relevant past insights based on a new prompt.

### Phase 2 (Qualitative Review):

A "Limited Expert Review" with 3–5 peers to evaluate the UI/UX flow and the emotional relevance of the AI's "mirroring" logic.

## Data Ingestion Sources:

**Primary Data:** User generated multimodal reflections (Text/Audio) processed via **VOSK**.

**Secondary Data:** Controlled synthetic datasets designed to evaluate the **Retrieval Augmented Generation (RAG)** pipeline.

## Key Performance Metrics:

**Semantic Precision:** Evaluation of the Cosine Similarity threshold to determine how accurately the AI identifies "dummy" historical insights.

**Orchestration Latency:** Measuring the end-to-end performance of the microservices (Auth -> Gateway -> AI Engine) during high-volume data ingestion.



# Project Plan

**Development Framework:** Agile (Scrum) Methodology

Project execution via 2 week iterative sprints to ensure modular integration.

## Phase 1: Foundation & Identity Infrastructure (Current Focus)

**Microservice Orchestration:** Initializing the **Eureka Discovery Server** and **Spring Cloud Gateway** for service registry and secure routing.

**Security Layer:** Implementing the **Auth Service** to manage user registration and stateless session handling via **JWT**.

**Frontend Scaffolding:** Developing the **Ionic/Angular** shell to establish connectivity between the presentation and backend layers.

## Phase 2: Multimodal Ingestion & Persistence

**Capture UI:** Building the primary reflection interface for mobile and web.

**Voice-to-Text:** Integrating the **VOSK** engine to perform local, privacy-focused transformation of audio into text.

**Semantic Storage:** Configuring **PostgreSQL with pgvector** to enable high-dimensional embedding storage for user memories.





# Project Plan

## Phase 3: Intelligence & Mirroring Engine

**Contextual Memory:** Developing the **RAG (Retrieval-Augmented Generation)** pipeline to link the vector database with the LLM.

**Pattern Synthesis:** Programming the **AI Engine** to perform semantic search and identify recurring emotional or behavioral themes.

**Proactive Mirroring:** Implementing the logic to "surface" past breakthroughs during new entries to reinforce user growth.

## Phase 4: Validation & Optimization

**Simulation Testing:** Stress-testing the RAG loop using a **Synthetic Dataset of 100+ "Aha!" moments** to measure Recall @ K accuracy.

**Performance Tuning:** Optimizing microservice latency and vector search speed.

**Final Deployment:** Production build of the **Android APK** and live hosting of the microservices ecosystem.



# References

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**Core Theory:** High-Dimensional Semantic Vector Space & Persistence.



# Thank You

Questions and Feedback

**Repository:** <https://github.com/prabhu-tejaa/MIRROR>  
**Web Interface:** <https://projectmirror.tech/>  
**Android Build:** <https://github.com/prabhu-tejaa/MIRROR/releases>

MIRROR

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